CLUSTER-WEIGHTED EDMD

Lorenzo Tomaz*, Judd Rosenblatt, Flavio Kicis, Thomas B. Jones, Diogo Schwerz de Lucena
AE Studio
{lorenzo, judd, flavio.kicis, thomas, diogo}@ae.studio

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Extended Dynamic Mode Decomposition (EDMD) is the canonical data-driven approximation of the Koopman operator [Williams et al., 2015a, Mezić, 2005, Brunton et al., 2022, Korda and Mezić, 2018, Mauroy et al., 2020]. Prior partitioning approaches predefine the partition by basin label [Williams et al., 2015a], phase-space stitching [Sinha et al., 2020], or operating-regime indicator [Peitz and Klus, 2019]. A complementary line of work enriches the global observable basis through learned dictionaries [Li et al., 2017], deep autoencoders [Lusch et al., 2018], or kernel methods [Williams et al., 2015b], orthogonal to the partitioning direction pursued here. We introduce *Cluster-Weighted EDMD* (CW-EDMD), which learns the partition jointly with per-cluster operators via Expectation-Maximization (EM) on a cluster-weighted-model joint density [Gershenfeld et al., 1999, Ingrassia et al., 2014, Punzo, 2014], with responsibilities combining geometric proximity and per-cluster prediction residual. Across three classical systems (36 configurations, 10 seeds), CW-EDMD improves over EDMD at the matched polynomial lift degree, including where EDMD itself saturates near machine precision.

Method. CW-EDMD is a Cluster-Weighted Model joint density

$$p(x_t, x_{t+1}) = \sum_{k=1}^{K} \pi_k \, \mathcal{N}(x_t; c_k, \Sigma_k) \, \mathcal{N}(\Delta x_k; 0, \sigma_k^2 I),$$

with per-cluster prediction residual $\Delta x_k = x_{t+1} - c_k - P_d \, K_k \, \Phi(x_t - c_k)$. Here Φ is a degree-q monomial lift centered at c_k, K_k the per-cluster Koopman matrix, and P_d the state projection. EM alternates between updating cluster responsibilities $r_{ik} \propto \pi_k \mathcal{N}(x_t^{(i)}; c_k, \Sigma_k) \mathcal{N}(\Delta x_k^{(i)}; 0, \sigma_k^2 I)$, which combine geometric proximity with per-cluster prediction residual, and solving each K_k in closed form by responsibility-weighted least squares, generalizing EDMD to the per-cluster regime (full derivation in Appendix A).

Experimental setup. Three classical systems: Lorenz (d=3, polynomial-RHS deg-3), damped pendulum (d=2, non-polynomial $\sin\theta$), double-well Duffing (d=2, polynomial-RHS deg-3). Each: 12 configurations sweeping five sampling distributions, data size, domain, integrator step, and fit budget; 10 fixed seeds per configuration (full setup in Appendix C; code and corpus in Appendix F). We test *matched-degree* CW-EDMD-q against EDMD-q via paired Wilcoxon tests; a *cross-config win* is $p < 0.05$ with lower CW-EDMD mean.

Table 1: Headline matched-degree summary: paired-test wins of CW-EDMD vs. EDMD over the 12 configurations per system, and median error ratio EDMD / CW-EDMD at one-step and 5 s rollout (> 1: CW-EDMD lower error).

	wins		ratio	
System (q, K)	1-step	5 s	1-step	5 s
Pendulum (4, 8)	11/12	10/12	76×	76×
Duffing (3, 8)	12/12	12/12	2.7×	2.4×
Lorenz (3, 12)	11/12	10/12	9.8×	23×

Results. CW-EDMD outperforms EDMD at the matched polynomial lift on all three systems (Table 1; per-system Pareto frontiers in Figures 1–3 and full tables in Appendix D). On Duffing the matched comparison spans three polynomial degrees ($q \in \{3, 4, 5\}$, all with $12/12$ wins, paired Wilcoxon $p < 10^{-4}$ throughout); the two 5 s rollout non-wins on Lorenz are characterized by the capacity-vs-data condition of Appendix B. A matched-(q, K) within-EDMD ablation across all three systems isolates the residual-aware mechanism: CW-EDMD wins $271/324$ paired comparisons (84%, Appendix E).

*Corresponding author.

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Appendix A: Full Method Derivation

Related work and positioning. The idea of partitioning phase space and fitting a separate Koopman operator per region is not new. The original EDMD paper [Williams et al., 2015a] demonstrates a partitioned EDMD on the Duffing oscillator by splitting the data along the basin-of-attraction boundary at $x{=}0$ and fitting separate operators to each basin, an early observation that per-region operators on dynamically meaningful partitions can outperform a single Koopman operator on the full phase space. Sinha, Nandanoori, and Yeung [Sinha et al., 2020] formalize phase-space stitching of region-specific Koopman operators for large-scale dynamical systems. Peitz and Klus [Peitz and Klus, 2019] develop a switched/multi-model Koopman formulation in which different operating regimes are assigned distinct linear Koopman models, with regime indicators driving the switch. Earlier mixture-of-experts work in the statistics literature [Jordan and Jacobs, 1994, Gershenfeld et al., 1999, Ingrassia et al., 2014, Punzo, 2014] provides the joint-density EM machinery that CW-EDMD reuses. The contribution of CW-EDMD relative to these works is the combination of three elements: (i) the partition is *learned jointly* with the per-cluster operators by EM rather than predefined by basin label, geometry, or operating-regime indicator; (ii) the responsibilities used in the expectation step combine geometric proximity with per-cluster prediction residual, so that the partition tracks where each operator predicts well rather than where data is geometrically dense; and (iii) the per-cluster predictor is the discrete EDMD operator on a recentered polynomial lift, giving a clean apples-to-apples baseline of CW-EDMD against EDMD at matched lift degree. The geometry-only ablation in Appendix E shows, on Duffing, that element (ii) is what drives the matched-degree advantage.

Two orthogonal axes of Koopman approximation. Beyond the partitioning literature surveyed above, a parallel line of work improves Koopman approximation by enriching the global observable basis: dictionary learning [Li et al., 2017], deep Koopman autoencoders [Lusch et al., 2018, Takeishi et al., 2017, Yeung et al., 2019], kernel EDMD [Williams et al., 2015b, Klus et al., 2020], and residual-based verification [Colbrook, 2023]. These methods retain a single global operator on a richer learned or kernel basis. CW-EDMD makes the orthogonal bet: a partitioned state space in which a simple local basis (recentered monomials) suffices per region. The two directions address different inductive failure modes — basis insufficiency for a single global operator versus operator insufficiency under a fixed simple basis — and compose naturally: a learned per-cluster basis is the obvious next step. For the matched-degree comparisons reported here, our baseline is therefore vanilla EDMD at the same monomial lift, which isolates the partitioning contribution; head-to-head comparison against learned-basis methods is left for future work, where per-cluster learned bases are the natural target.

CWM joint density. Given paired training data $\{(x_t^{(i)}, x_{t+1}^{(i)})\}_{i=1}^N$ in $\mathbb{R}^d$, we model the joint density as

$$p(x_t, x_{t+1}) = \sum_{k=1}^{K} \pi_k \, p_X(x_t \mid k) \, p_{Y|X}(x_{t+1} \mid x_t, k),$$

where $p_X(x_t \mid k) = \mathcal{N}(x_t;\, c_k, \Sigma_k)$ is the cluster's geometric component and $p_{Y|X}(x_{t+1} \mid x_t, k) = \mathcal{N}(\Delta x_k;\, 0, \sigma_k^2 I)$ is the per-cluster residual density, with $\Delta x_k = x_{t+1} - \hat{x}_{t+1|k}$ the prediction error of cluster k on the observed transition. Mixture weights π_k satisfy $\sum_k \pi_k = 1$. Compared with a Gaussian mixture model (GMM), the second factor introduces *residual awareness*: cluster k takes responsibility only for points it can predict well, not just points geometrically nearby.

Per-cluster Koopman operator. For each cluster k with center c_k, define the recentered monomial lift $\Phi : \mathbb{R}^d \to \mathbb{R}^{M_q}$ of total degree $\le q$, where $M_q = \binom{d+q}{q}$. We use the bias-first EDMD convention with the constant monomial in entry 1 followed by the d linear monomials, $\Phi(z) = [1,\, z_1,\, \ldots,\, z_d,\, z_1^2,\, z_1 z_2,\, \ldots]^\top$, so that the linear monomials $z_1, \ldots, z_d$ occupy entries $2, \ldots, d+1$. The projection $P_d \in \mathbb{R}^{d \times M_q}$ defined by $P_d = [0_{d\times 1} \mid I_d \mid 0_{d\times(M_q-d-1)}]$ then satisfies $P_d \Phi(z) = z$. With $K_k \in \mathbb{R}^{M_q \times M_q}$ the per-cluster Koopman matrix, the per-cluster prediction is

$$\hat{x}_{t+1|k} = c_k + P_d \, K_k \, \Phi(x_t - c_k),$$

We retain the full square K_k rather than a rectangular $\mathbb{R}^{d\times M_q}$ regression onto the state because the square form is the standard EDMD discretization of the Koopman operator on the lifted observable space [Williams et al., 2015a, Brunton et al., 2022]: it preserves the full lifted-space spectrum and lets the operator be reused for spectral analyses, eigenfunction extraction, and downstream control. The output projection P_d is applied only at prediction time, not at fit time, so the optimisation (below) targets the full M_q-dimensional lifted residual rather than only the state-projected residual. The cost is the $M_q^2 - d\,M_q$ unused parameters per cluster; we report this overhead in the parameter counts of all tables.

EM updates. Let r_{ik} be the posterior responsibility of cluster k for sample i. The expectation step (responsibility update) is

$$r_{ik} \propto \pi_k \, \mathcal{N}(x_t^{(i)}; c_k, \Sigma_k) \, \mathcal{N}\big(\Delta x_k^{(i)}; 0, \sigma_k^2 I\big),$$

combining geometric proximity (first Gaussian) with prediction residual (second Gaussian). The maximization step (parameter update) then updates each cluster:

$$c_k = \frac{\sum_i r_{ik} x_t^{(i)}}{\sum_i r_{ik}}, \quad \Sigma_k = \frac{\sum_i r_{ik} (x_t^{(i)} - c_k)(x_t^{(i)} - c_k)^\top}{\sum_i r_{ik}},$$

$$K_k = \arg \min_{K \in \mathbb{R}^{M_q \times M_q}} \sum_{i=1}^{N} r_{ik} \left\| \Phi(x_{t+1}^{(i)} - c_k) - K\, \Phi(x_t^{(i)} - c_k) \right\|_2^2.$$

Stack the lifted training pairs as $\tilde{X}_k, \tilde{Y}_k \in \mathbb{R}^{M_q \times N}$ (i-th column of $\tilde{X}_k$ is $\Phi(x_t^{(i)} - c_k)$, i-th column of $\tilde{Y}_k$ is $\Phi(x_{t+1}^{(i)} - c_k)$) and let $W_k = \mathrm{diag}(r_{1k}, \ldots, r_{Nk}) \in \mathbb{R}^{N \times N}$. The objective then reads $J(K) = \mathrm{tr}\big[(\tilde{Y}_k - K\tilde{X}_k)\, W_k\, (\tilde{Y}_k - K\tilde{X}_k)^\top\big]$. Setting $\partial J / \partial K = 0$ gives $\tilde{Y}_k W_k \tilde{X}_k^\top = K\, \tilde{X}_k W_k \tilde{X}_k^\top$, whose minimum-norm solution is

$$K_k = \big(\tilde{Y}_k W_k \tilde{X}_k^\top\big)\big(\tilde{X}_k W_k \tilde{X}_k^\top\big)^\dagger \in \mathbb{R}^{M_q \times M_q},$$

where $(\cdot)^\dagger$ denotes the Moore–Penrose pseudoinverse, applied only to the right factor. We solve this via SVD-based weighted least squares, which handles rank-deficient lifted Grams through SVD truncation. Finally σ_k^2 is updated from the responsibility-weighted residual variance.

Empty-cluster pruning. A weak Dirichlet prior $\mathrm{Dir}(\alpha)$ on $\boldsymbol{\pi}$ with $\alpha < 1$ allows components with vanishing responsibility to be pruned between iterations.

Initialization. We initialize c_k by K-means and K_k by ordinary EDMD on the points initially assigned to cluster k. EM is run with multiple restarts and the highest-likelihood run is retained.

Rollout. At inference, given current state x_t, the rollout cluster index is $k^*(x_t) = \arg\max_k \pi_k\, \mathcal{N}(x_t; c_k, \Sigma_k)$ (geometry-only at test time, since no x_{t+1} is available), and the next state is $x_{t+1} = c_{k^*} + P_d\, K_{k^*}\, \Phi(x_t - c_{k^*})$. Multi-step rollouts iterate this map.

Appendix B: Capacity-vs-Data Operating Regime

Each cluster's Koopman operator $K_k \in \mathbb{R}^{M_q \times M_q}$ has M_q^2 free parameters, with $M_q = \binom{d+q}{q}$. EM with K clusters distributes the N training pairs across components, giving roughly P/K samples per cluster after responsibility assignment, where P is the count of *decorrelated* training samples. Under the classical regression-underdetermination heuristic that parameter count should not exceed roughly half the sample count, this motivates the candidate condition

$$K \cdot M_q^2 \leq \tfrac{1}{2}\, P_{\text{eff}}.$$

Post-hoc nature of the rule. We are explicit that this condition is *post-hoc*: it was derived after observing the two Lorenz 5 s rollout non-wins, by asking what those configurations had in common. It is not a prospective prediction validated against held-out failures. The rule has tunable degrees of freedom (the one-half safety factor; the choice of timescale τ entering P_{eff}; and the choice of $P_{\text{eff}} \approx N\, \Delta t / \tau$ as the proxy for decorrelated sample count, in the absence of a closed-form expression for chaotic dynamics), so a fit between the rule and the observed failures is necessary but not sufficient evidence for predictive validity. What we report is consistency: the two Lorenz 5 s non-wins occur in regimes where KM_q^2 approaches or exceeds P_{eff}, while the ten winning Lorenz regimes have P_{eff} several times larger than KM_q^2, and damped pendulum and Duffing (smaller M_q, denser data) are uniformly well below the underdetermination threshold. Prospective validation, on a new system where the rule predicts a failure that is then observed, is the natural next test; more principled estimators of P_{eff} based on autocorrelation length or block-bootstrap effective sample size would also tighten the rule. For Lorenz at q=3, $M_q^2 = 400$. With N=500 and K=12, one has $K \cdot M_q^2 = 4800 \gg 250$, an order-of-magnitude violation: this is the small-training-data Lorenz attractor configuration. With $\Delta t = 0.005$ on the chaotic Lorenz attractor, P_{eff} at N=2000 collapses far below 2000 due to high temporal correlation between consecutive states, again violating the rule: this is the short-integrator-step Lorenz configuration. All 10 winning Lorenz configurations satisfy $P_{\text{eff}} \geq 4M_q^2$ comfortably. The rule is restrictive but predictive: it tells the practitioner when partitioning is worth the parameter cost, and is the principal methodological constraint we report.

Appendix C: Full Experimental Setup

Systems. The three classical systems used in this work, all expressed as continuous-time autonomous ODEs and then integrated to produce discrete-time pairs (x_t, x_{t+1}) at fixed step Δt:

Lorenz attractor ($d = 3$, polynomial RHS of degree 2; the bilinear terms xy, xz are the highest-degree nonlinearity):

$$\dot{x} = \sigma(y - x), \quad \dot{y} = x(\rho - z) - y, \quad \dot{z} = xy - \beta z,$$

with parameters $(\sigma, \rho, \beta) = (10, 28, 8/3)$. The attractor exhibits strange-attractor chaos with two folding wings around the unstable origin.

Damped pendulum ($d = 2$, non-polynomial RHS):

$$\dot{\theta} = \omega, \qquad \dot{\omega} = -\sin\theta - \gamma\,\omega,$$

with damping coefficient $\gamma = 0.2$. The non-polynomial $\sin\theta$ term is the reason no global polynomial lift is exact at any finite degree; this is the system on which the parameter-efficiency advantage of CW-EDMD is largest.

Double-well Duffing oscillator ($d = 2$, polynomial RHS of degree 3):

$$\dot{x} = v, \qquad \dot{v} = x - x^3 - \delta\,v,$$

with damping $\delta = 0.25$. The cubic restoring force $x - x^3$ creates two stable foci at $x = \pm 1$ and an unstable saddle at $x = 0$. The RHS polynomial degree is 3; in our experiments the EDMD lift degree q=5 is where global EDMD saturates near machine precision, with q=3 matching the RHS degree itself.

Discretization. All systems are integrated with vectorized RK4. Pair generation produces (x_t, x_{t+1}) at fixed step Δt, with $\Delta t \in \{0.005, 0.01, 0.05, 0.1\}$ across configurations.

Sampling distributions. Five distributions are swept independently per system: *uniform* over a configurable box, *Gaussian* centered at the origin, *Gaussian mixture* centered at attractor foci, *periodic-noise* (sinusoidal with additive noise), and *trajectory ensemble* (random ICs integrated for several steps, all trajectory points used as training). Additional Lorenz-specific *single-trajectory attractor* sampling is used for two attractor-baseline configurations.

Configurations. 12 configurations per system, each varying one factor relative to a baseline (data size, box size, Δt, fit budget, sampling distribution).

Seeds. 10 fixed seeds per (system, configuration, method): $\{1, 42, 101, 307, 1001, 7789, 13245, 11, 103, 13\}$. Seeds control train/test sampling, EM initialization, and integrator noise.

Methods. EDMD at degrees $\{2, 3, 4, 5\}$ (Lorenz, damped pendulum, Duffing) and $\{6, 8\}$ (damped pendulum only); CW-EDMD at degrees $\{2, 3\}$ (Lorenz), $\{2, 4\}$ (damped pendulum), $\{2, 3, 4, 5\}$ (Duffing) with $K \in \{2, 4, 8, 16\}$ where the capacity-vs-data rule permits.

Metrics. (i) *One-step error*: mean ℓ_2 prediction error on the held-out test set. (ii) *Rollout error at H seconds*: mean ℓ_2 error of the iterated map at horizons $H \in \{1, 2, 5, 10, 20\}$ s, averaged over test initial conditions.

Statistics. For each (system, configuration, metric) we run all methods over 10 seeds and compute mean and 95% CI. *Paired Wilcoxon signed-rank tests* (per-seed pairing) test the matched-degree comparison "CW-EDMD q vs. EDMD q". A cross-configuration outcome is a *win* if $p < 0.05$ and the CW-EDMD mean is lower; a *loss* if $p < 0.05$ and higher; a *tie* otherwise. We report wins / losses / ties tallied across the 12 configurations.

Appendix D: Per-System Detailed Results

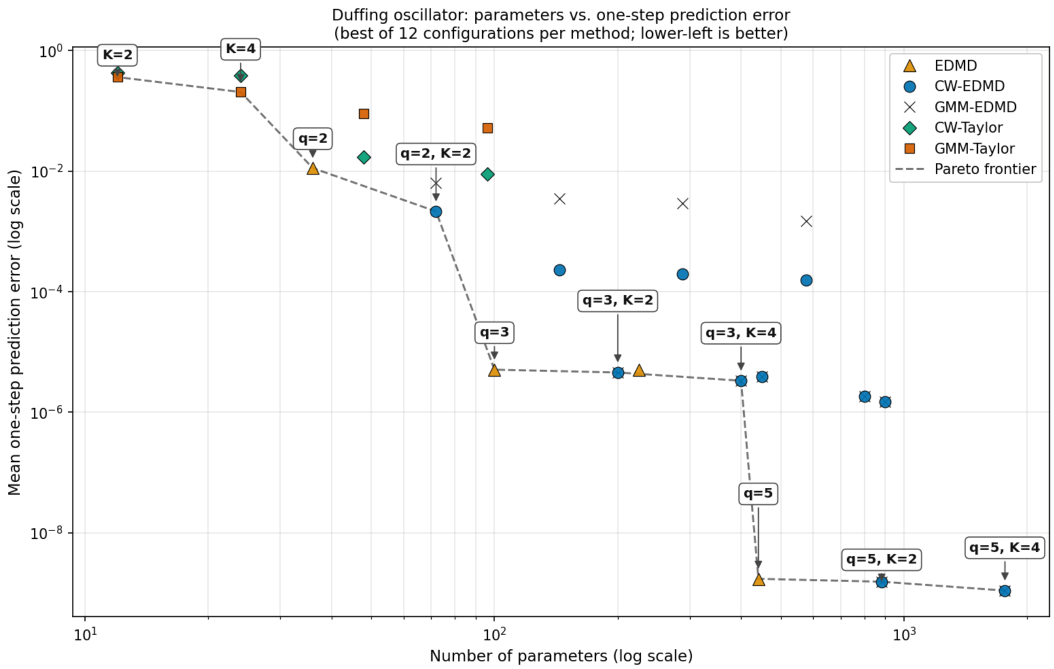

Figure 1: **Duffing Pareto frontier: median one-step error vs. parameter count.** Each point is one (method, hyperparameter) combination summarized across 12 configurations $\times$ 10 seeds. Ablation variants below (GMM-EDMD, CW-Taylor, GMM-Taylor) are formally defined in Appendix E. Blue: CW-EDMD (the focus of this paper). Orange: EDMD at increasing polynomial degree. Brown: GMM-EDMD (Appendix E, the within-EDMD ablation with the residual-aware responsibility update replaced by GMM-style proximity-only responsibilities). Green: CW-Taylor (Appendix E, the Taylor-predictor variant). Pink: GMM-Taylor (Appendix E, the within-Taylor ablation: geometry-only responsibilities + per-cluster regression, the standard CWM contrast point). Dashed: Pareto frontier. CW-EDMD dominates EDMD at the RHS-matched lift (q=3) and at the saturating-EDMD lift (q=5, machine-precision regime).

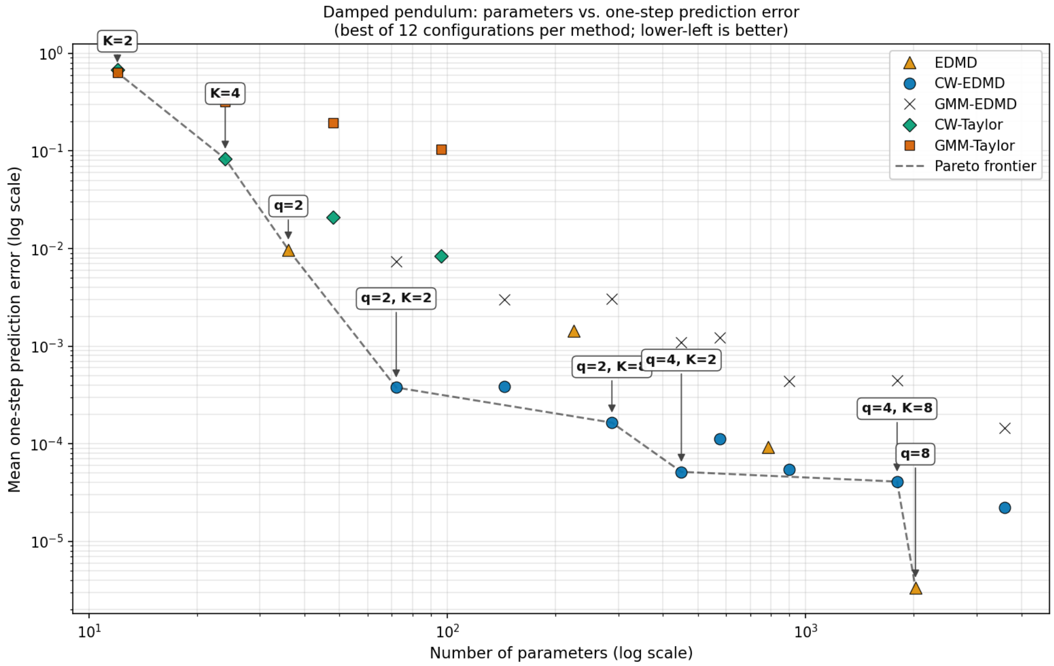

Figure 2: **Damped pendulum Pareto frontier: median one-step error vs. parameter count.** Each point is one (method, hyperparameter) combination at its best-case configuration over 12 configurations $\times$ 10 seeds. Ablation variants (GMM-EDMD, CW-Taylor, GMM-Taylor) are formally defined in Appendix E. Blue: CW-EDMD (the focus of this paper). Orange: EDMD at increasing polynomial degree. Brown: GMM-EDMD (Appendix E, the within-EDMD ablation with the residual-aware responsibility update replaced by GMM-style proximity-only responsibilities). Green: CW-Taylor (Appendix E, the Taylor-predictor variant). Pink: GMM-Taylor (Appendix E, the within-Taylor ablation: geometry-only responsibilities + per-cluster regression, the standard CWM contrast point). Dashed: Pareto frontier. Matched-degree CW-EDMD (q=4, blue stars at the bottom of the plot) strictly dominates all EDMD parameter budgets, including q=8 at 2025 parameters. Higher polynomial degrees overfit; partitioning does not.

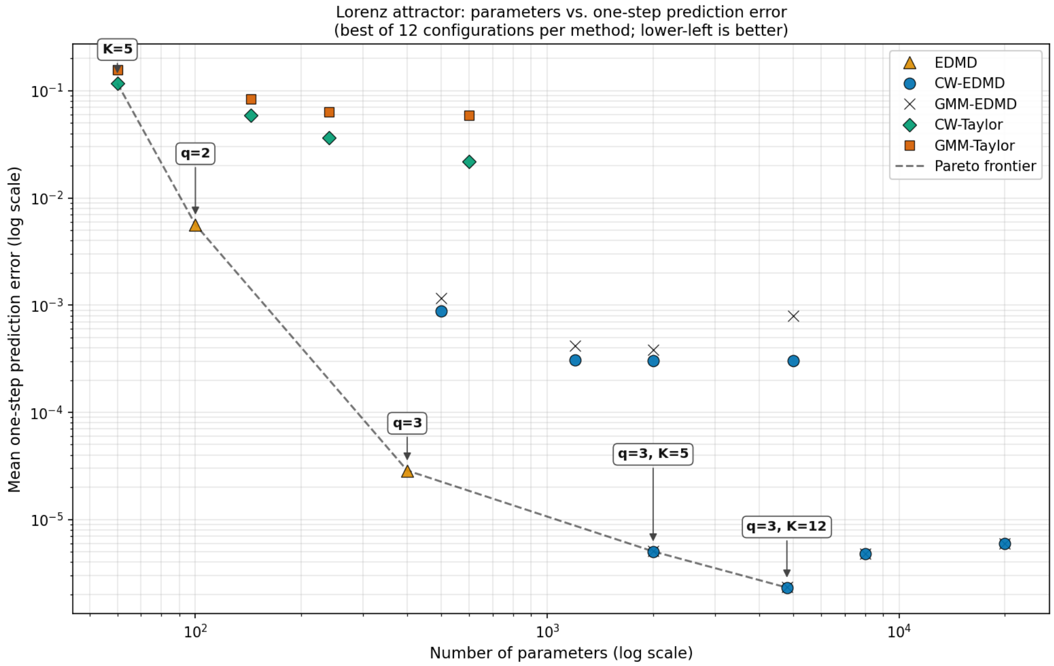

Figure 3: **Lorenz Pareto frontier: median one-step error vs. parameter count.** Each point is one (method, hyperparameter) combination at its best-case configuration over 12 configurations × 10 seeds. Ablation variants (GMM-EDMD, CW-Taylor, GMM-Taylor) are formally defined in Appendix E. Blue: CW-EDMD (the focus of this paper). Orange: EDMD at increasing polynomial degree. Brown: GMM-EDMD (Appendix E, the within-EDMD ablation with the residual-aware responsibility update replaced by GMM-style proximity-only responsibilities). Green: CW-Taylor (Appendix E, the Taylor-predictor variant). Pink: GMM-Taylor (Appendix E, the within-Taylor ablation: geometry-only responsibilities + per-cluster regression, the standard CWM contrast point). Dashed: Pareto frontier. CW-EDMD at the matched lift degree q=3 Pareto-dominates EDMD on the ten configurations satisfying the capacity-vs-data constraint of Appendix B. The two non-winning configurations (small training data and short integrator step) are dominated by the operating-regime issue, not the partitioning principle.

Damped pendulum. At matched lift degree q=4, median prediction error in dimensionless state-space units across the 12 configurations and 10 seeds is reported in Table 2. Median is reported in preference to mean because rollout error on a small subset of high-dynamic-range configurations diverges to large values that dominate the cross-configuration mean.

Table 2: Damped pendulum at matched lift degree q=4: median prediction error (dimensionless state-space units) across 12 configurations and 10 seeds. Bold marks the column-wise minimum.

Method	Params	one-step	5 s
EDMD q=2	36	2.5×10^{-2}	1.1×10^{0}
EDMD q=4	225	4.6×10^{-3}	3.1×10^{-1}
CW-EDMD q=4, K=4	900	8.5×10^{-5}	5.0×10^{-3}
CW-EDMD q=4, K=8	1800	6.1×10^{-5}	4.1×10^{-3}
CW-EDMD q=4, K=16	3600	$\mathbf{5.6\times10^{-5}}$	$\mathbf{2.5\times10^{-3}}$

At matched lift degree q=4, CW-EDMD K=8 wins on $11/12$ configurations on one-step and $10/12$ on 5 s rollout against EDMD q=4. The K=16 row reduces 5 s rollout error further but raises one-step error only marginally, consistent with mild one-step overfitting at the larger cluster count that still generalises slightly better over the 5 s horizon.

Duffing. At three matched polynomial degrees, median prediction error across the 12 configurations and 10 seeds is reported in Table 3, with EDMD baselines at every matched degree for direct comparison.

Table 3: Duffing oscillator: median prediction error (dimensionless state-space units) across 12 configurations and 10 seeds. Top block: EDMD at four polynomial lift degrees. Bottom block: CW-EDMD at three matched lift degrees. Bold marks the column-wise minimum.

Method	Params	one-step	5 s
EDMD q=2	36	5.2×10^{-2}	7.7×10^{-1}
EDMD q=3 (matches RHS deg)	100	6.5×10^{-5}	6.6×10^{-3}
EDMD q=4 (higher lift)	225	6.5×10^{-5}	6.6×10^{-3}
EDMD q=5 (higher lift)	441	6.8×10^{-8}	4.6×10^{-6}
CW-EDMD q=3, K=8	800	2.4×10^{-5}	2.7×10^{-3}
CW-EDMD q=4, K=4	900	1.7×10^{-5}	1.3×10^{-3}
CW-EDMD q=5, K=4	1764	$\mathbf{4.1\times10^{-8}}$	$\mathbf{3.1\times10^{-6}}$

CW-EDMD wins on all twelve configurations at each of the three matched degrees against EDMD ($p < 10^{-4}$ in every configuration). The within-Taylor ablation against the standard GMM-clustered local model contrast point of the CWM literature is reported in Appendix E.

Lorenz. Median prediction error across the 12 configurations and 10 seeds is reported in Table 4. The metric *r5s* reaches the rollout-step cap on the chaotic configurations and saturates; *one-step* is the cleanest cross-method comparison.

Table 4: Lorenz attractor: median prediction error (dimensionless state-space units) across 12 configurations and 10 seeds. Top block: EDMD at two polynomial lift degrees. Middle block: CW-EDMD at lift degree q=2 (mismatched). Bottom block: CW-EDMD at the matched lift degree q=3. Bold marks the column-wise minimum.

Method	Params	one-step	5 s
EDMD q=2	100	3.8×10^{-2}	1.5×10^{1}
EDMD q=3	400	1.6×10^{-3}	2.4×10^{0}
CW-EDMD q=2, K=5	500	3.3×10^{-3}	7.0×10^{0}
CW-EDMD q=2, K=12	1200	6.4×10^{-4}	1.4×10^{0}
CW-EDMD q=2, K=20	2000	4.6×10^{-4}	7.7×10^{-1}
CW-EDMD q=3, K=5	2000	1.7×10^{-4}	1.3×10^{-1}
CW-EDMD q=3, K=12	4800	$\mathbf{1.6\times10^{-4}}$	1.1×10^{-1}
CW-EDMD q=3, K=20	8000	1.7×10^{-4}	$\mathbf{9.4\times10^{-2}}$

At the matched lift degree q=3, CW-EDMD with K=12 wins on ten of twelve configurations on 5 s rollout against EDMD q=3 (paired Wilcoxon $p < 0.05$); the remaining two configurations correspond to capacity-vs-data violations described in Appendix B (small training data and short integrator step). At fixed K=5 (500–2000 parameters), increasing the lift degree from q=2 to q=3 decreases one-step error by roughly two orders of magnitude, demonstrating that the matched lift degree is essential for the partitioning advantage on this polynomial-RHS system. The mismatched q=2 CW-EDMD entries are included for ablation.

Appendix E: Generality and Initialization

CWM framing and approximator-agnostic per-cluster fit. CW-EDMD instantiates the cluster-weighted-model framework [Gershenfeld et al., 1999, Ingrassia et al., 2014, Punzo, 2014] with EDMD as the per-cluster predictor. The CWM joint density $p(x_t, x_{t+1}) = \sum_k \pi_k\, p_k(x_t)\, p_k(x_{t+1} \mid x_t)$ is agnostic in the per-cluster predictor $p_k(x_{t+1} \mid x_t)$. We provide a Taylor-expansion variant in our implementation that replaces the per-cluster EDMD operator with a first-order centered linearization $\hat{x}_{t+1|k} = x_t + \Delta t\,[f(c_k) + J(c_k)(x_t - c_k)]$ (one explicit-Euler step on the linearized vector field) when analytic f and J are available at cluster centers; the same EM machinery applies. This drop-in substitutability is what we mean by *model-agnostic*: any local approximator that admits a fit-time loss and an inference-time prediction can be substituted. The headline matched-degree comparisons in the main text are restricted to EDMD by design, to make the partitioning hypothesis a clean controlled test on one approximator family.

Within-EDMD ablation: CW-EDMD vs. GMM-clustered EDMD at matched (q, K). To isolate the contribution of the residual-aware responsibility update from the partitioning effect itself, we compare CW-EDMD against an ablation in which the residual factor is removed from the E-step (the responsibility update in Appendix A, *EM updates*). Concretely, the CW-EDMD responsibility update

$$r_{ik} \propto \pi_k \mathcal{N}(x_t^{(i)}; c_k, \Sigma_k) \mathcal{N}(\Delta x_k^{(i)}; 0, \sigma_k^2 I)$$

is replaced by the geometry-only GMM-style update

$$r_{ik} \propto \pi_k \mathcal{N}(x_t^{(i)}; c_k, \Sigma_k),$$

which is the $\sigma_k^2 \to \infty$ limit of the residual factor (the residual likelihood degenerates to a constant and drops out). The M-step (updates to c_k, Σ_k, K_k, π_k) is held identical, and the per-cluster EDMD predictor and lift degree q are held fixed; the *only* difference between the two variants is the presence or absence of the residual factor in the E-step. We denote this variant *GMM-EDMD*. It is not a standard baseline in the EDMD literature; it is the natural within-family ablation that isolates the residual-aware responsibility mechanism. The ablation is run on all three systems at every configured (q, K) pair, with paired Wilcoxon tests per configuration. Across 324 paired comparisons (system $\times$ q $\times$ K $\times$ configuration), CW-EDMD wins on 271 (84%) at $p < 0.05$ with lower mean error, ties on 49 (15%), and loses on 4 (1.2%).

The losses concentrate in a single failure mode: Lorenz at $q{=}3, K{=}50$, where the capacity-vs-data condition of Appendix B is violated by an order of magnitude ($KM_q^2 = 20{,}000$ parameters against on the order of 10^3 effective samples per typical configuration). The same condition explains the within-EDMD outcome: when each cluster's K_k is severely under-determined, the per-cluster residuals Δx_k are dominated by fitting noise rather than partition signal, so CW-EDMD's residual-aware E-step trusts a noisy quantity and over-fits it. GMM-EDMD's geometry-only E-step ignores the residuals and is therefore more robust in this regime: the residual-aware update is a feature only when the residuals it consumes carry usable signal. The ties concentrate in another characterizable regime: polynomial-RHS systems at the lift degrees where EDMD itself saturates near machine precision (e.g., on Duffing at $q{=}3, K{=}8$, CW-EDMD median $2.4{\times}10^{-5}$ vs. GMM-EDMD median $2.5{\times}10^{-5}$; both methods land at the same floor and the residual-aware advantage is detectable but practically nil). The wins concentrate where the lift cannot fully capture the local dynamics: the damped pendulum at every tested (q, K) (96/96 wins, factors of 1.7×–3,100×), and polynomial-RHS systems at under-exact lift (Lorenz at $q{=}2$: $42/0/6$, factors 1.7×–3.4×; Duffing at $q{=}2$: $46/1/1$, factors 7.5×–53×). Within the same EDMD predictor family, then, the residual-aware update contributes *decisively* when the polynomial lift is mismatched to the local dynamics, and *negligibly* when the lift is sufficient and both responsibility-update strategies converge on the analytical floor. This is the direct mechanistic measurement that closes the inference: residual-awareness is the source of CW-EDMD's gain in regimes where the partition has work to do, and partitioning per se carries the gain in regimes where the lift suffices.

Two-mechanism decomposition of CW-EDMD's advantage. The within-EDMD ablation also implies a decomposition of CW-EDMD's gain over global EDMD into two contributing mechanisms: (i) *partitioning*, measurable as GMM-EDMD vs. EDMD (a per-cluster EDMD operator on a geometrically-clustered partition beats a single global operator at matched lift), and (ii) *residual-aware responsibilities*, measurable as CW-EDMD vs. GMM-EDMD. On polynomial-RHS systems at the lift where EDMD saturates (Lorenz $q{=}3$, Duffing $q{\geq}3$), mechanism (i) carries the gain (e.g., Lorenz $q{=}3, K{=}12$: GMM-EDMD already 40× better than EDMD, with CW-EDMD adding nothing further). On the damped pendulum at any q and on polynomial-RHS systems at lower lift, mechanism (ii) dominates (pendulum $q{=}4, K{=}8$: GMM-EDMD only 1.2× better than EDMD, while CW-EDMD is roughly 2,500× better than GMM-EDMD).

Cross-predictor confirmation: within-Taylor ablation. The same ablation in the Taylor predictor branch (CW-Taylor vs. GMM-Taylor, the classical CWM-vs-two-stage contrast point in the statistical-clustering literature [Ingrassia et al., 2014]) shows the same qualitative pattern, with the residual-aware mechanism contributing decisively at moderate-to-large K on all three systems. The agreement across predictor families confirms the mechanism is intrinsic to the responsibility-update form, not specific to the EDMD predictor.

Initialization. We initialize c_k by K-means and K_k by ordinary EDMD on the points initially assigned to cluster k, with multi-restart EM. Multi-restart is essential on Duffing wide-box configurations where the cubic nonlinearity creates deep local minima.

Computational cost. EDMD fits a single closed-form regression of size M_q^2 in one shot. CW-EDMD replaces this with K regressions of the same size per EM iteration, with T iterations and R restarts, giving a fit cost of order $R \cdot T \cdot K \cdot M_q^2 \cdot N$ versus $M_q^2 \cdot N$ for EDMD. In the configurations reported here ($T \leq 80$ iterations, $R \leq 2$ restarts,

$K \leq 16$), the worst-case CW-EDMD fit is roughly $80 \cdot 2 \cdot 16 \approx 2.6 \times 10^3$ times the cost of a single EDMD fit. Inference cost is comparable to EDMD: rollout requires one $K_k\, \Phi(\cdot)$ matrix-vector product per step plus a Gaussian-likelihood evaluation to pick the active cluster.

Limitations. Three limitations of the present study are worth flagging. First, the operating-regime condition of Appendix B is post-hoc: it was derived to be consistent with the two observed Lorenz 5 s rollout non-wins, not pre-registered or validated on held-out failures, and the proxy $P_{\text{eff}} \approx N\, \Delta t/\tau$ has tunable degrees of freedom. The condition should be treated as a hypothesis to be tested prospectively on a new system where it predicts a failure that is then observed, rather than as a validated bound. Second, the Lorenz results are mixed (ten wins, two non-wins out of twelve configurations on 5 s rollout against EDMD), and while the non-wins are characterizable in terms of capacity-vs-data, the absence of a clean win across all Lorenz configurations is a real limitation of the cross-system claim. Third, the corpus is restricted to autonomous low-dimensional systems with smooth dynamics; control-input-driven systems and higher-dimensional flows (where CW-EDMD's parameter count grows quadratically in M_q) are out of scope here.

Appendix F: Reproducibility

All experimental configurations are versioned as YAML files dispatched through a single statistical-validation driver (`validation/run_statistical.py -config <name>.yaml`). Per-seed JSON results are written for every (system, configuration, method, seed) combination and aggregated post-hoc into the long-form CSV used for the figures and tables in this paper. The full corpus (39,564 observations across systems / configurations / methods / seeds / metrics) and code are available at `https://github.com/agencyenterprise/fractal_residual_aware_clustering`.